

ED-FEED: An AI-Driven Intelligent Student Feedback Analysis and Insight Generation System

Evaluation of the TensorFlow Model

In this phase of the project, the sentiment analysis system was evaluated after upgrading the model from a traditional machine learning approach to a deep learning-based TensorFlow model using a Bidirectional Long Short-Term Memory (BiLSTM) architecture. The goal of this improvement was to enhance the system's ability to understand contextual and sequential patterns in student feedback data. The dataset was prepared by selecting the *comments* column as the input feature and *sentiment_label* as the target variable. Data preprocessing involved handling missing values by replacing them with empty strings and converting all entries into string format to ensure compatibility with the model. The text data was then transformed into numerical representations suitable for deep learning, allowing the model to process sequences of words effectively.

The BiLSTM model was designed to capture both forward and backward contextual dependencies in the text, enabling it to better understand sentence structure and meaning compared to earlier approaches. During evaluation, the model demonstrated improved performance in classifying student feedback into positive, neutral, and negative sentiments. Unlike the previous Logistic Regression model, which relied primarily on word frequency and simple patterns, the BiLSTM model was able to learn deeper semantic relationships within the data. This allowed it to handle complex sentence structures, mixed sentiments, and subtle variations in language more effectively. As a result, the model showed better accuracy and more balanced performance across all sentiment classes.

The evaluation results also indicated a reduction in misclassification, particularly between neutral and positive classes, which were challenging for the earlier model. The BiLSTM model provided more consistent predictions and improved generalization across different types of feedback. Its ability to process word sequences helped it interpret the context of the sentences more accurately, making it suitable for real-world text analysis tasks.

Overall, the transition to a TensorFlow-based BiLSTM model significantly improved the sentiment analysis system. The model's enhanced ability to capture contextual and sequential information resulted in more accurate and reliable predictions. This evaluation highlights the effectiveness of deep learning techniques in natural language processing and demonstrates that advanced models like BiLSTM are better suited for analyzing complex and unstructured student feedback data compared to traditional machine learning methods.

RESULT

Classification Report

Class	Precision	Recall	F1-Score	Support
Negative	0.83	0.90	0.86	11807
Neutral	0.81	0.82	0.81	11807
Positive	0.88	0.79	0.84	11807

Overall Performance

Metric	Value
Accuracy	0.84
Macro Avg F1	0.84
Weighted Avg F1	0.84

The model achieved an overall accuracy of 84%, indicating good performance in correctly classifying student feedback.

Confusion Matrix

Actual \ Predicted	Negative	Neutral	Positive
Negative	10574	917	316
Neutral	1166	9713	928
Positive	1010	1430	9367